

MATH 6112. FALL 2026. ASSIGNMENT 1.

DUE: SEPTEMBER 16, 11:59 PM

Instructions.

- (i) You must return your homework typed on letter size paper. Text must be typed in 10pt font or larger; this page is typed in 12pt fonts, but the “Version heading” on the top right is made in 10pt fonts.
- (ii) The homework must be uploaded in the directory “Assignments” on Canvas by midnight (EDT) of the due day. Submission of a “pdf” file is required.
- (iii) Please, keep your arguments short. You do not need to rederive things we have done in class, and can just refer to the class notes. If in doubt, please ask.
- (iv) Careful and precise mathematical writing is important. Arguments which are hard to follow will be marked down.
- (v) The Homework must reflect your own work. You are welcome to clarify the problems with one another and with me, but in the end you need to work on your own. Use of any source different from the class notes must be explicitly mentioned. This includes use of any AI tool, like **Chatgpt** or the like.

For questions, W 3:30-5, or e-mail, or by appointment.

Homework 1 Problems.

Unless otherwise specified, all transformations, matrices, vectors, etc., are over an arbitrary field $\mathbb{F}$. You can use all you need from Chapter 1 of the Lecture Notes.

(1) [15 points.] Using only the material developed up to that point, do Exercise 1.3.13-(1, 2, 6).

(2) [15 points.] Prove Theorem 1.5.10. (This is the one about similar matrices corresponding to the same linear transformation represented in different bases).

(3) [15 points.] Here, the field is $\mathbb{C}$. Let $A \in \mathbb{C}^{n \times n}$ have distinct eigenvalues $\lambda_1, \dots, \lambda_p$, of algebraic multiplicities $m_1, \dots, m_p$, respectively.

(a) Show that for each $k = 1, \dots, p$, the solution set of

$$(0.1) \quad (A - \lambda_k I)^{m_k} v = 0$$

is spanned by m_k linearly independent vectors.

(b) Show that putting together (for $k = 1, \dots, p$) the sets of vectors from (0.1) gives a basis for $\mathbb{C}^n$.

(4) [15 points.] This exercise lays down a technique which is considered a pretty safe way to obtain the Jordan form. In fact below we build a matrix V which gives us a certain Jordan form of a matrix $A \in \mathbb{C}^{n \times n}$. The idea is to assume that there exists V : $V^{-1}AV = J$, where J is in Jordan form, and see how we can get V .

(a) First, suppose that A has only one distinct eigenvalue, and one Jordan block

of size n : $V^{-1}AV = \lambda I + \begin{bmatrix} 0 & 1 & \dots & 0 \\ \vdots & \ddots & \ddots & \vdots \\ 0 & \dots & 0 & 1 \\ 0 & \dots & 0 & 0 \end{bmatrix}$. Explain how we could obtain V

in this case.

(b) Generalize part (a) to the case of A having a Jordan form with several Jordan blocks.

(5) [10 points.] Let $J \in \mathbb{C}^{n \times n}$ be a Jordan block of size n relative to the eigenvalue λ . Show this formula for the entries of the matrix J^k , $k \geq 1$:

$$(J^k)_{ij} = \begin{cases} \binom{k}{j-i} \lambda^{k-(j-i)} , & i \leq j \leq i+k , \\ 0 & \text{otherwise} . \end{cases}$$

(6) [15 points.] Suppose we have a (n, n) differentiable complex matrix valued function of one parameter μ , real or complex, which we indicate as $\mu \in \mathbb{F}$. That is, we have a function $A : \mu \in \mathbb{F} \rightarrow \mathbb{C}^{n \times n}$, such that all of the entries of A are differentiable function of μ . Write A column-wise: $A = [a_1 \ a_2 \ \cdots \ a_n]$. Show that

$$\frac{d(\det A(\mu))}{d\mu} = \det(a'_1, a_2, \dots, a_n) + \det(a_1, a'_2, \dots, a_n) + \cdots + \det(a_1, a_2, \dots, a'_n) ,$$

where a'_i denotes derivative of a_i with respect to μ .

(7) [15 points.] Exercise 1.5.46, that is the problem about uniqueness of block-diagonal form.